

MediKiosk

AI-Powered Clinical History & Document Intelligence Platform

Problem Statement ID: 26047
Patient Case-Taking Software

Organization: Ministry of AYUSH | Department: All India Institute of Ayurveda
Category: Software | Theme: MedTech / BioTech / HealthTech

1. Problem Statement

1.1 Background

Clinical history-taking — systematically eliciting a patient's chief complaint, history of present illness, past medical and surgical history, drug and allergy history, family and personal history, and a review of systems — is widely regarded as the single most valuable diagnostic activity in medicine. A well-conducted history alone is reported to yield the correct diagnosis in 70-80% of cases, before any physical examination or investigation is performed.

India's public hospital outpatient departments (OPDs), however, operate under extreme patient load. Tertiary and apex government hospitals routinely register between 4,000 and 10,000 patients per day, compressing average doctor-patient consultation time to just 2-5 minutes — among the shortest in the world (a 2017 BMJ Open study spanning 67 countries placed India's average primary-care consultation at just over two minutes). Within this narrow window, the physician must elicit history, examine the patient, review prior records, diagnose, counsel, and prescribe — all at once. History-taking is invariably the first casualty of this time pressure, resulting in incomplete histories, missed comorbidities, repeated questioning across visits, and avoidable diagnostic error.

1.2 The AYUSH-Specific Dimension

The problem is compounded in Ayurvedic OPDs. Ayurvedic diagnosis requires a far more extensive assessment framework than allopathic intake — Trividha, Ashtavidha, and Dashavidha Pariksha — covering Prakriti (constitution), Vikriti (current imbalance), Agni (digestive capacity), Koshta (bowel nature), Ahara-Vihara (diet and lifestyle), Nidana (causative factors), and Samprapti (pathogenesis). Capturing this depth of assessment manually within a 2-5 minute OPD slot is effectively impossible, forcing practitioners to abbreviate the very process that defines personalised Ayurvedic care.

1.3 The Records Fragmentation Problem

Compounding the time constraint is the state of patient records. Patients typically carry physical, unstructured documents — prescriptions, lab reports, discharge summaries, imaging films — from multiple prior providers, often handwritten, in varying languages, and chronologically disordered. The physician must manually scan through this pile during the same limited consultation window, further reducing time available for direct patient interaction.

India's national digital health backbone, the Ayushman Bharat Digital Mission (ABDM), provides ABHA health IDs and FHIR-based interoperability standards for exchanging health records. However, a 'first-mile' gap remains: no patient-facing platform currently captures structured clinical history and digitises physical documents before the patient enters the consultation room and feeds this into the ABDM ecosystem.

1.4 Why Existing Approaches Fall Short

- Hospital registration kiosks capture only demographic/appointment data (name, age, department, token) — no clinical content whatsoever.
- Mobile health apps and tele-triage chatbots require smartphone literacy, stable connectivity, and pre-visit enrolment — excluding the elderly, rural, low-literacy, and first-visit patients who form the bulk of government OPD load.

- Manual nurse-led triage desks are human-resource-limited and cannot scale to 5,000+ daily patients without reintroducing the same time bottleneck.
- Generic document scanners digitise images only — they do not extract, structure, or chronologically organise clinical content, nor link it to ABHA or a structured history.

1.5 Precise Problem Definition

There is no purpose-built, patient-facing software platform that enables patients to independently and comprehensively record their medical history — through both natural spoken conversation and guided touchscreen interaction — while simultaneously digitising their existing physical medical documents, to generate a structured, physician-ready clinical history summary that integrates with the hospital information system (HIS) and the ABDM ecosystem before the patient enters the consultation room.

1.6 Core Challenges to Overcome

- Multilingual, multi-accent voice capture in noisy hospital environments (Hindi, English, and major regional languages), for patients with varying literacy and digital comfort.
- Accessibility for elderly and low-literacy users through icon-driven UI, audio prompts, and conversational guidance — usable with zero prior training.
- Accurate clinical structuring — converting free-form narration into a standardised, physician-readable history, including AYUSH-specific Dashavidha Pariksha parameters.
- Reliable OCR of handwritten and printed medical documents across languages, with intelligent extraction of diagnoses, medications, and investigation values.
- Privacy, consent, and security compliance with the Digital Personal Data Protection Act (DPDP) 2023 and the ABDM consent framework.

2. Proposed Solution — MediKiosk

MediKiosk is an AI-powered clinical history software platform that allows any patient — regardless of literacy, language, or digital familiarity — to complete a comprehensive medical history interview and digitise their existing paper records before ever entering the consultation room. The output is a structured, physician-ready summary pushed automatically to the Hospital Information System (HIS) and linked to the patient's ABHA record.

2.1 Design Principles

- Zero-training usability — a first-time, non-tech-savvy patient must be able to complete the flow unaided.
- Dual-mode interaction — every question answerable by voice or by tapping, so no single ability (literacy, hearing, comfort with technology) becomes a blocker.
- Physician-in-the-loop — AI drafts, never diagnoses. Every summary is editable and must be confirmed by the physician before being saved to the record.
- Safety-first — emergency symptoms are detected and escalated immediately, bypassing the normal queue.
- Privacy-by-design — explicit, revocable, audio-explained consent; encrypted processing; no retained data beyond what is authorised.

2.2 Solution Modules

Module A — Conversational Multimodal History Engine

Conducts a structured clinical interview via voice and touch simultaneously. Indian-language ASR converts speech to text; a dialogue manager constrained by a clinical history ontology asks adaptive follow-up questions — e.g. on hearing 'chest pain', it probes onset, character, radiation, and aggravating/relieving factors following the SOCRATES framework used by physicians. An AYUSH mode extends the interview to capture the ten Dashavidha Pariksha parameters. A red-flag detector continuously scans responses for emergency indicators (e.g. chest pain with breathlessness, stroke symptoms) and triggers an immediate priority alert to triage staff.

Module B — Medical Document Digitisation & Intelligence

Patients scan prior prescriptions, lab reports, and discharge summaries through the kiosk camera. A document-AI pipeline performs high-accuracy OCR (printed and handwritten, multilingual), extracts diagnoses, medications with dosages, and investigation values, arranges documents into a chronological timeline, and flags out-of-range lab values or potential drug interactions.

Module C — Structured History Summary Generator

An LLM-based summarisation layer merges the conversational history and digitised documents into one concise, standard-format clinical summary (Chief Complaint → HPI → Past History → Drug/Allergy → Family → Personal → Review of Systems → Prior Investigations), displayed on the physician's screen the moment the patient enters the room. The physician can edit, confirm, or reject any part before it is saved — the AI output is always a draft, never an autonomous diagnosis.

Module D — Consent, Privacy & ABDM Integration

Patients authenticate via ABHA ID and grant granular, revocable, audio-explained consent. The structured history is pushed to the hospital HIS/EMR and linked to the patient's ABHA Personal Health Record via FHIR

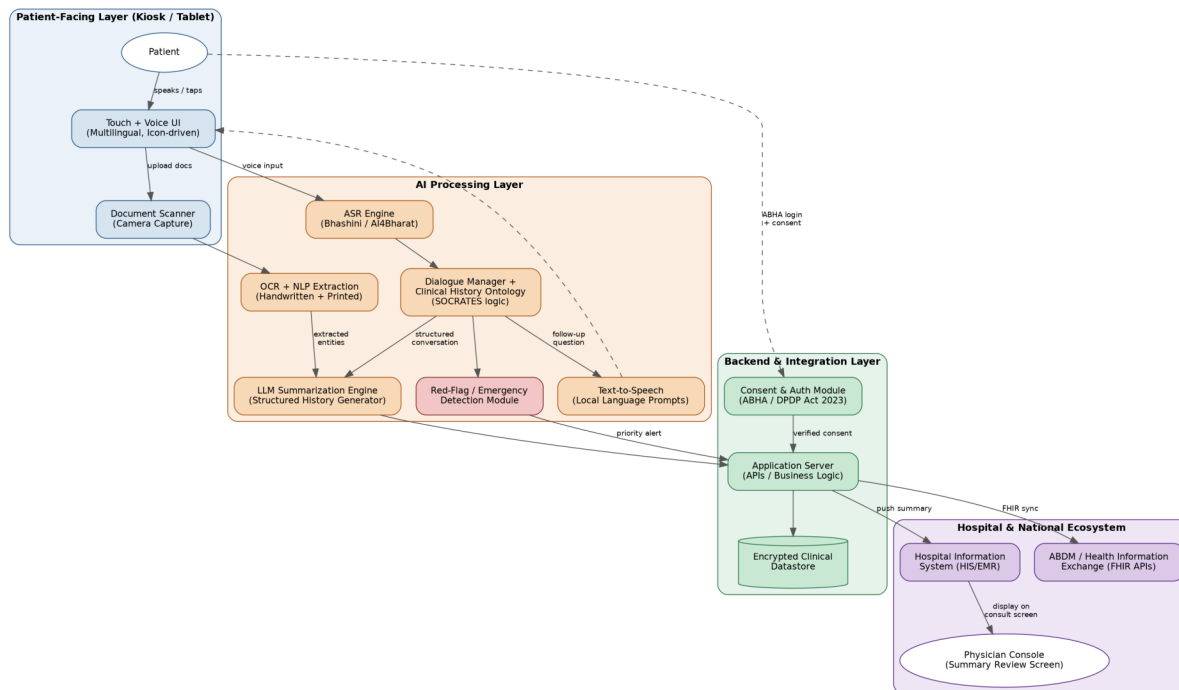
APIs. Session data is processed securely and cleared immediately after submission, in compliance with the DPDP Act 2023.

2.3 End-to-End Patient Journey

- Step 1 — Identify: Patient scans/enters ABHA ID or registers as new; selects preferred language; grants audio-guided consent.
- Step 2 — Converse: AI conducts an adaptive voice + touch interview covering chief complaint, HPI, and full history; red flags trigger immediate priority triage.
- Step 3 — Scan: Patient uploads prior prescriptions, lab reports, and discharge summaries; the system digitises, structures, and timelines them.
- Step 4 — Summarise & Route: AI generates the structured summary, links it to ABHA, and pushes it to the HIS ahead of the consultation.
- Step 5 — Consult: The physician reviews the complete history in seconds, edits/confirms it, and spends the full consultation on examination, reasoning, and counselling.

3. System Architecture

The architecture is organised into four layers: a patient-facing kiosk layer, an AI processing layer, a backend/integration layer, and the hospital/national ecosystem layer it connects into. Data flows from patient input through AI structuring into a physician-reviewable record, with consent and security controls gating every hand-off.



3.1 Layer Description

- **Patient-Facing Layer:** Touchscreen kiosk/tablet running the dual-mode (voice + touch) interface and document scanner/camera capture.
- **AI Processing Layer:** ASR, dialogue manager with clinical ontology, text-to-speech, OCR/NLP extraction, red-flag detection, and the LLM summarisation engine.
- **Backend & Integration Layer:** Application server (APIs and business logic), encrypted clinical datastore, and the consent/authentication module.
- **Hospital & National Ecosystem Layer:** Hospital Information System (HIS/EMR), ABDM Health Information Exchange (via FHIR APIs), and the physician's consultation-room console.

4. Technology Stack

Layer	Component	Suggested Technology
Frontend (Kiosk)	Touch + voice UI	React / Flutter (tablet-optimised), multilingual icon-driven UX
Frontend (Kiosk)	Audio capture & playback	WebRTC / native audio APIs
AI — Speech	ASR (speech-to-text)	Bhashini / AI4Bharat Indic ASR models
AI — Speech	Text-to-speech	Bhashini TTS / Coqui TTS (Indic voices)
AI — Conversation	Dialogue management	LLM (e.g. fine-tuned open-weight model) + clinical ontology / rules engine
AI — Documents	OCR (printed + handwritten)	Google Cloud Vision / Tesseract + custom handwriting model
AI — Documents	Clinical entity extraction	Medical NLP (spaCy/clinical BERT variants) for diagnoses, drugs, values
AI — Summarisation	Structured summary generation	LLM with structured-output (JSON/FHIR-aligned) prompting
Backend	Application server	Node.js / Python (FastAPI) microservices
Backend	Database	PostgreSQL (structured records) + object storage for scanned documents
Backend	Consent & Auth	ABHA OAuth integration, DPDP-compliant consent ledger
Integration	Interoperability	HL7 FHIR APIs for ABDM Health Information Exchange
Integration	HIS connectivity	HL7/FHIR or hospital-specific HIS adapter APIs
Infrastructure	Deployment	On-prem or empanelled cloud (data-residency compliant), containerised (Docker/Kubernetes)
Security	Data protection	End-to-end encryption, role-based access control, audit logging

5. Expected Impact & Feasibility

5.1 Impact

- Frees consultation time for examination, clinical reasoning, and patient counselling instead of manual history-taking.
- Reduces diagnostic error from incomplete or rushed histories.
- Enables AYUSH practitioners to capture Dashavidha Pariksha depth previously sacrificed to time pressure.
- Extends digital health record coverage to elderly, rural, and low-literacy populations who are excluded by app-based alternatives.
- Closes the ABDM 'first-mile' gap by feeding structured, patient-originated data into the national health data ecosystem.

5.2 Feasibility

The solution is feasible today because its core enabling technologies have matured independently: Indian-language ASR (Bhashini/AI4Bharat), general-purpose LLMs capable of structured clinical summarisation, high-accuracy OCR for handwritten medical documents, and ABDM's published FHIR interoperability standards. MediKiosk's contribution is the integration of these components into a single, consent-governed, patient-facing workflow — not the invention of new underlying AI capability.

5.3 Safeguards

- The AI never issues a diagnosis; all output is a draft requiring physician confirmation.
- All consent is granular, revocable, and explained audibly for low-literacy users.
- Session data is cleared immediately after submission; only the physician-confirmed summary is persisted.